

exp 1 case 1.sci (C:\Users\pachi\Documents\exp 1 case 1.sci) - SciNotes

exp 1 case 1.sci

```
1 clc;
2 clear;
3
4 // Given values
5 x = 3;
6 y = 4;
7 z = -6;
8
9 // Cartesian to Cylindrical Coordinates
10 r = sqrt(x^2 + y^2);
11 phi = atan(y, x) * 180 / %pi; // Angle in degrees
12
13 disp("Cartesian to Cylindrical Coordinates (r, phi, z):");
14 disp([r, phi, z]);
15
16 // Cartesian to Spherical Coordinates
17 R = sqrt(x^2 + y^2 + z^2);
18 theta = acos(z / R) * 180 / %pi; // Polar angle in degrees
19 phi = atan(y, x) * 180 / %pi; // Azimuth angle in degrees
20
21 disp("Cartesian to Spherical Coordinates (R, theta, phi):");
22 disp([R, theta, phi]);
23
```

Scilab 2025.1.0 Console

File Edit Control Applications ?



Scilab 2025.1.0 Console

```
"Cartesian to Cylindrical Coordinates (r, phi, z):"
5.  53.130102 -6.
"Cartesian to Spherical Coordinates (R, theta, phi):"
7.8102497  140.19443  53.130102
```

--> |

exp:- 1

$$x = 3$$

$$y = 4$$

$$z = -6$$

$$r = \sqrt{3^2 + 4^2} = \sqrt{25} = 5$$

$$\phi = \tan^{-1}\left(\frac{4}{3}\right) = 53.1$$

$$z = -6$$

$C \rightarrow R$

$$x = r \cos \phi$$

$$y = r \sin \phi$$

$$R = \sqrt{3^2 + 4^2 + (-6)^2} = \sqrt{61} \approx 7.810$$

$$\phi = \cos^{-1}(-6/7.810) \approx 140.1^\circ$$

$$\theta = 53.13$$

o/p ✓
In ✓

$$0.600$$

$$0.799$$